Daniel Jang

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Technical Skills

Languages: Java, C, C++, Python, JavaScript, HTML/CSS, TypeScript, SQL, Ruby, Dart
Developer Tools: Git, VS Code, Linux, Docker, GCP, Bash, Maven, Powershell, ArcGIS, Cursor
Frameworks: React, Node.js, Spring Boot, Flutter, Rails, Flask, Vite, Gemini API

Projects

Scout! – Indoor Navigation for the Blind | *Kotlin, React Native, Expo, ONNX Runtime, YOLO* Sep. 2026

- Built an infrastructure-free indoor navigation prototype for visually impaired users, with the rear camera as a real-time perception layer requiring no sensors.
- Designed a three-layer on-device pipeline: compressed JPEG capture, **YOLO** object detection via **ONNX Runtime** and **Google ML Kit** OCR for sign reading, with **Gemini Flash** as a fallback when on-device vision was uncertain and slow.
- Placed **3rd at the hackathon**, delivering a working prototype using **ARCore** that reads signs, flags trip hazards, and speaks directions without any venue hardware.

Siel – Digital Biographer | *Java 17, Spring Boot, Node.js, PowerShell, Gemini API* May 2026

- Designed a photo-driven journaling platform to encourage self-reflection through modern technologies, using in-memory processing to extract metadata without committing sensitive media to disk, eliminating the local data footprint.
- Developed a high-concurrency metadata pipeline in **Java 17** using ParallelStreams to execute multi-threaded EXIF extraction, significantly reducing processing time for bulk image batches.
- Hardened system security via Argon2id hashing and RS256 asymmetric **JSON Web Token** signing, integrating the Gemini API to generate journal narratives within a secure environment.

Mappin' – Destination Alarm System | *TypeScript, JavaScript, HTML, CSS* Apr. 2026

- Constructed a cross-platform location alarm to notify users upon arrival at their intended destinations by implementing Service Worker logic between a Chrome Extension (MV3) and a **Progressive Web app**.
- Optimized data extraction by implementing a URL parser that uses URL APIs to resolve shortened redirects, ensuring successful destination mapping across **30+ varying coordinates** with a **0% failure rate** for common share types.
- Implemented a high-precision arrival detection system by integrating the Haversine formula, allowing the model to calculate real-time optimal distance and trigger synchronized haptic alerts across user-defined radii from **50m** to **2000m**.

Experience & Activities

BA Consulting Group Ltd Sep. 2026 – Present

Geospatial Web App Developer Co-op *Toronto, ON*

- Developed full-stack platform features using **Node.js**, **SQL**, and REST APIs, enabling 100+ engineers and planners to query, manipulate, and manage large-scale transportation datasets across Canadian provinces.
- Built dynamic geospatial visualization modules using **Mapbox GL JS** and **ArcGIS**, rendering multi-layered spatial data to analyze traffic patterns and transit metrics in real time.
- Hardened web application security by securing REST endpoints, standardizing input validation, and restructuring API key management to eliminate client-side credential exposure.

WATonomous May 2026 – Aug. 2026

Website Developer *Waterloo, ON*

- Improved the web experience by including project showcases and interactive graphic designs using **React** and **TypeScript** to display the team's autonomous and robotic innovations.
- Optimized platform performance and code maintainability by designing front-end components with **Tailwind CSS** and fixing site bugs.
- Expanded website content, implementing dynamic blogging features and collaborating with the marketing subgroup to drive student recruitment.

Waterloo Rocketry Nov. 2025 – Aug. 2026

Software Member *Waterloo, ON*

- Analyzed data collection workflows in **Python** and **Java** to ensure software stability during rocket launch operations.
- Collaborated with the team to develop a full-stack ground control software using **TypeScript** and **Python**.

Education

University of Waterloo Waterloo, ON

Bachelor of Computer Science, Honours | Term Distinction *Sep. 2025 – Apr. 2030*

- **Relevant Coursework:** Algorithms, Memory Management, Data Structures, Imperative, Functional & Object-Oriented Programming, Shell Scripting, Linear Algebra I, Calculus II, Probability, Graph Theory.